

Nonuniform-Sampling-Enhanced LSTM-FCN Hybrid Network for Inverse Design of Optically Transparent Ultrawideband AFSR

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Abstract—A new modeling framework integrating Ramer–Douglas–Peucker (RDP) nonuniform sampling (NUS) with a long short-term memory (LSTM)-fully connected network (FCN) hybrid neural network (LFN) is proposed for the inverse design of a single-layer optically transparent ultrawideband (UWB) absorptive frequency-selective reflector (AFSR). Unlike conventional uniform sampling (US), the RDP-NUS accurately extracts key features while efficiently reducing overall data volume without additional data cropping or dimensionality reduction algorithms. Comparative studies demonstrate that the proposed RDP-LFN framework consistently outperforms US-FCN approaches, achieving lower loss with sparse sampled points and various FCN depths. Utilizing the RDP-LFN framework, a single-layer optically transparent UWB AFSR is designed. The measurement of the fabricated prototype exhibits good agreement with numerical simulation, validating the efficacy and feasibility of the proposed method in addressing the challenging requirements for multifunctional electromagnetic devices in modern electronic systems. This is the first attempt to introduce a NUS strategy into neural networks enabled FSS design, offering new insights for intelligent design of microwave devices.

Index Terms—Absorptive frequency-selective reflector, long short-term memory-fully connected network hybrid neural network, nonuniform sampling, optically transparent, Ramer–Douglas–Peucker algorithm.

I. INTRODUCTION

IN MODERN radar systems, electromagnetic (EM) wave manipulation is crucial for enhancing equipment survivability and operational effectiveness. Compared to bandstop frequency-selective surfaces (FSS) [1], absorptive FSSs (AFSSs) [2], and frequency selective rasorbers (FSRs) [3], the absorptive frequency-selective reflector (AFSR), capable of comprehensively reflecting radar signals while absorbing adversarial jamming frequencies, significantly enhances electronic

countermeasure capabilities and has emerged as a prominent research focus. However, the lossy layers in existing AFSRs predominantly employ lumped-resistor-loaded metallic FSS [4], [5], [6], [7], [8], [9], [10], which limit applications in scenarios where optical transparency is required.

Conventional FSS design relies on time-intensive simulations and empirical tuning. To address this, neural networks (NNs) have been introduced in FSS designs for powerful feature extraction and nonlinear mapping capabilities. Researchers have successfully achieved forward prediction, as well as forward and inverse designs of diverse structures, including reflective [11], [12], [13], [14], [15]; transmissive [16], [17], [18], [19], [20], [21], [22], [23]; absorptive [24], [25], [26], [27], [28]; absorption-transmission-integrated [29], [30]; and polarization-converting [31], [32] FSSs. However, limited research effort has been reported in the public domain concerning NN-assisted AFSR design.

Particularly, in NN-assisted FSS inverse design, data pre-processing directly while significantly impacts the computational cost and model performance. Existing studies have approached this problem through two main lines of investigation: *dataset reduction*, such as evolutionary algorithm-assisted process [18], equivalent-circuit guidance [26], and vector-graph-feature-extraction [27]; *dimensionality reduction*, including Fourier transform [23], and VAE compression [32], and transfer-function fitting [33]. However, these methods highly depend on uniform sampling (US) strategy of EM responses (e.g., *S*-parameters). Frequency-based US requires a dense dataset for multifunctional FSS designs, whose *S*-parameters typically exhibit multiple peaks and troughs. If the salient features of the EM response can be captured with a limited number of nonuniform sampled (NUS) points, dimensionality reduction would no longer require additional algorithms. Similarly, dataset reduction could be complemented by data compression, further improving training efficiency. This inspires the idea of magnitude-based sampling.

Hence, a novel FSS inverse design framework that combines Ramer–Douglas–Peucker (RDP)-NUS with a long short-term memory (LSTM)-fully connected network (FCN) hybrid neural network (LFN) is proposed. Unlike conventional US-FCN approaches, the proposed RDP-LFN framework significantly reduces both data reconstruction error and training loss. To validate its effectiveness, we designed and experimentally realized

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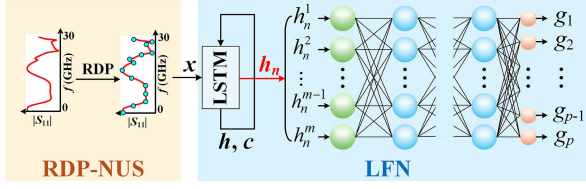
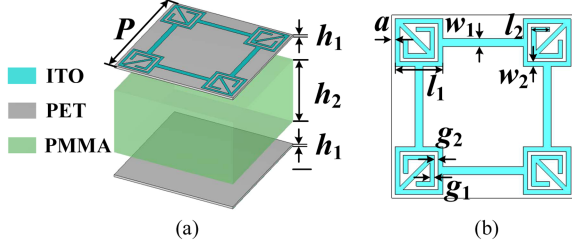


Fig. 1. Architecture of the proposed RDP-LFN framework.


 Fig. 2. Unit cell topology of the proposed AFSR: (a) perspective and (b) top views. ($P = 10$, $h_1 = 0.175$, $a = 0.2$, $g_1 = 0.25$. Unit: mm.)

a single-layer optically transparent UWB AFSR, exhibiting a good agreement between simulated and measured results. This is the first attempt to introduce a NUS strategy into NN-assisted inverse FSS design, offering a new alternative for intelligent design of microwave devices.

II. PROPOSED METHOD

To effectively reduce data dimensionality while maintaining FSS inverse design accuracy, a novel RDP-LFN model is proposed, as illustrated in Fig. 1. First, the framework employs the RDP algorithm to perform NUS on the original $|S_{11}|$ parameter curves, achieving significant data compression while preserving critical EM response characteristics. The sampled data x is then fed into the LFN, where the LSTM captures frequency-dependent temporal correlations, and the FCN extracts global spectral features. Through concatenation of both modules, the network generates p -dimensional geometric parameters, thereby completing the end-to-end inverse design process from EM response to geometric structure.

The RDP algorithm, a classical curve simplification technique, removes redundant points based on a fixed distance threshold [34] and has been widely applied in graphics processing and geospatial mapping. However, employing a fixed threshold results in inconsistent numbers of sampled points across different curves, which is not feasible for training an NN due to the requirement of consistent input dimensions. To address this issue, a fixed-number RDP sampling method is developed, wherein a binary search iteratively adjusts the threshold for each curve until exactly N points are sampled. To the best of our knowledge, this is the first attempt to introduce an RDP-NUS strategy into NN-FSS design. During each iteration, the threshold is dynamically updated depending on whether the current number of sampled points exceeds or falls below N , and the process repeats until convergence is achieved. The complete procedure is summarized in Algorithm 1, taking the $|S_{11}|$ curve of the structure shown in Fig. 2 as an illustrative example.

To evaluate the feature extraction capability of the RDP algorithm, Fig. 3 presents the cumulative reconstruction error

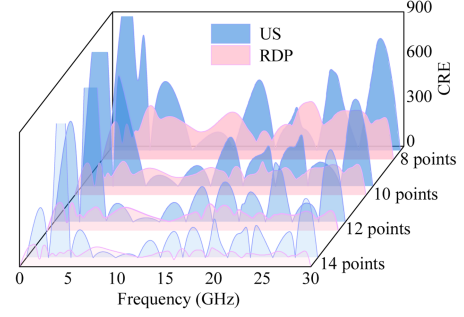


Fig. 3. CRE for US and RDP algorithms at each frequency point with respect to sampled points of 8, 10, 12, and 14.

Algorithm 1: RDP-Based Nonuniform Sampling.

Require: A curve $|S_{11}| = \{p_1, p_2, \dots, p_{1001}\}$; Desired sampling point count N ; Infinitesimal ς (e.g., $\varsigma = 10^{-10}$)
Ensure: Sampled curve $|S_{11}'| = \{p_1, p_2, \dots, p_N\}$

- 1: Initialize $\xi_{\text{low}} \leftarrow 0$, $\xi_{\text{high}} \leftarrow [\max(|S_{11}|) - \min(|S_{11}|)]$
- 2: **while** $(\xi_{\text{high}} - \xi_{\text{low}}) > \varsigma$ **do**
- 3: $\xi_{\text{mid}} \leftarrow (\xi_{\text{high}} + \xi_{\text{low}}) / 2$
- 4: $|S_{11}'| \leftarrow \text{RDP}(|S_{11}|, \xi_{\text{mid}})$ [34]
- 5: **if** $\text{len}(|S_{11}'|) < N$ **do**
- 6: update $\xi_{\text{high}} \leftarrow \xi_{\text{mid}}$
- 7: **elif** $\text{len}(|S_{11}'|) > N$ **do**
- 8: update $\xi_{\text{low}} \leftarrow \xi_{\text{mid}}$
- 9: **else return** $|S_{11}'|$
- 10: **end while**
- 11: **return** $|S_{11}'| \leftarrow \text{RDP}(|S_{11}|, \xi_{\text{high}})$

(CRE) comparison between RDP and US for various numbers of sampled points (8/10/12/14 points). The CRE at frequency f is defined as: $\sum_i ||S_{11}^i| - |S_{11}^i|*|$, where i is the curve index of the dataset for the AFSR structure in Fig. 2, $|S_{11}|$ is the original reflection coefficient magnitude, and $|S_{11}|*$ is the reconstructed reflection coefficient after linear interpolation of $|S_{11}'|$. The CRE decreases with the increase of sampling frequency points, as shown in Fig. 3. Compared to US, the RDP algorithm achieves lower mean CRE by 20%, 41%, 50%, and 45% for 8, 10, 12, and 14 sampled points, respectively, demonstrating its enhanced capability in capturing key features of $|S_{11}|$ curves. Consequently, the RDP-NUS provides higher-quality datasets for subsequent LFN training.

Given that the LSTM algorithm is inherently applicable to processing non-uniform data [35], this work employs its gating mechanisms to analyze RDP-sampled $|S_{11}|$. The short-term memory cells of LSTM can effectively capture localized abrupt variations in $|S_{11}|$, while the long-term memory cells model cumulative effects across the entire spectrum. This dual-scale feature extraction mechanism is critical for accurate learning of absorption-reflection responses of AFSRs.

III. DATASET PREPARATION AND LFN TRAINING

A. Dataset Preparation

To validate the inverse design capability of the proposed LFN, a transparent AFSR structure is exploited as illustrated in Fig. 2.

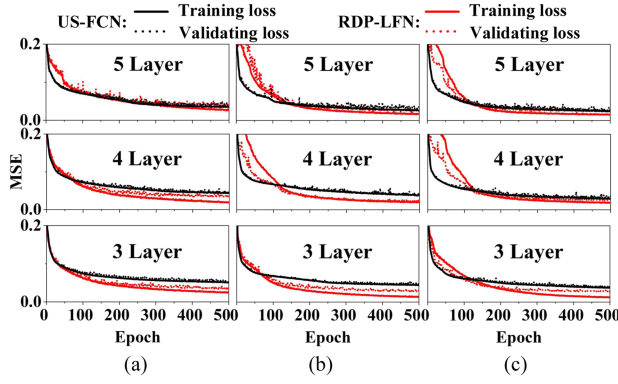


Fig. 4. Comparison of training and validation losses between US-FCN and RDP-LFN for sampled points of (a) 8, (b) 10, and (c) 14.

The structure consists of three vertically stacked layers: a ground plane, a dielectric spacer, and an FSS layer. The FSS and ground employ ITO resistive films with sheet resistances of $12 \Omega/\text{sq}$ and $5 \Omega/\text{sq}$, respectively, patterned on polyethylene terephthalate (PET) substrates ($\epsilon_r = 3$, $\text{tg}\delta = 0.06$). The FSS geometry was elaborately designed to achieve a wide operational bandwidth and polarization insensitivity while maintaining sufficient design flexibility. The transparent dielectric spacer utilizes poly-methyl methacrylate (PMMA, $\epsilon_r = 2.25$, $\text{tg}\delta = 0.0002$).

By comprehensively considering the mutual constraints among geometric parameters, fabrication feasibility, and overall performance requirements, the specific parameter ranges are determined as follows in unit of mm: $h_2 = [5: 0.5: 5.5]$, $g_2 = [0.2: 0.1: 0.5]$, $l_1 = [3: 0.3: 4.5]$, $l_2 = [0.5: 0.5: 2]$, $w_1 = [0.4: 0.4: 2.4]$, $w_2 = [0.2: 0.1: 0.5]$. A total of 4608 $|S_{11}|$ responses were obtained using CST across the 0–30 GHz frequency range. The dataset was partitioned into training, testing, and validation sets at an 8:1:1 ratio. These data are processed as follows: a) perform RDP sampling on all $|S_{11}|$ responses; and b) calculate the frequency differences Δf between adjacent sampled points and form data pairs x by combining with corresponding $|S_{11}|$ values: $x_j = \{f_1, |S_{11}|^1 (j = 1); \Delta f_j, |S_{11}|^j (j > 1)\}$.

The Δf as an additional input feature enhances the LSTM's adaptive weighting capability for critical frequency bands. The 64-dimensional hidden state output from the final timestep of the LSTM is processed through FCN layers with ReLU activation functions to further extract global information, ultimately yielding the 6-dimensional geometric parameters.

B. Results and Discussion

The network is implemented in PyTorch, employing a mean squared error (MSE) loss function and Adam optimizer, with a batch size of 64 for 500 epochs. Fig. 4 presents the training and validation loss variations of the RDP-LFN model versus the US-FCN model, both employing 3-, 4-, and 5-layer FCN configurations for sampled points of 8, 10, and 14, corresponding to hidden layers with $\{128\}$, $\{128, 32\}$, and $\{128, 64, 32\}$ neurons, respectively. It can be observed that with 5 layers and only 8 sampled points, the limited information combined with a relatively deep architecture results in mild overfitting [36], and RDP-LFN exhibits only a modest performance advantage.

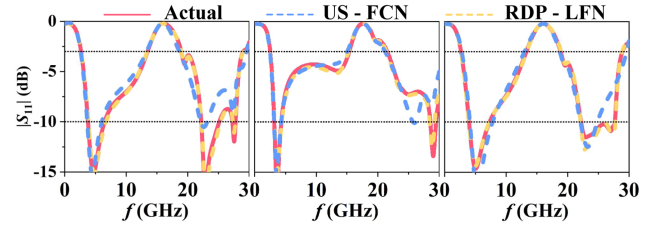


Fig. 5. Comparison between the actual $|S_{11}|$ and the simulated $|S_{11}|$ based on the inversely predicted structural parameters by US-FCN and RDP-LFN.

TABLE I
COMPARISON OF TRAINING, VALIDATION, AND TEST LOSSES ACROSS DIFFERENT FCN DEPTHS AND SAMPLED POINTS

Depths	Points	Method	Train	Validate	Test
3	8	US-FCN	0.0511	0.0551	0.0512
		FFT-FCN	0.0780	0.0611	0.0589
		PCA-FCN	0.0803	0.0822	0.0792
		RDP-LFN	0.0254	0.0357	0.0364
	14	US-FCN	0.0379	0.0407	0.0375
		FFT-FCN	0.0468	0.0517	0.0523
5	8	PCA-FCN	0.0513	0.0582	0.0528
		RDP-LFN	0.0132	0.0283	0.0246
	14	US-FCN	0.0350	0.0397	0.0340
		FFT-FCN	0.0348	0.0408	0.0379
	14	PCA-FCN	0.0626	0.0680	0.0632
		RDP-LFN	0.0270	0.0394	0.0325
5	8	US-FCN	0.0251	0.0287	0.0283
		FFT-FCN	0.0198	0.0266	0.0272
	14	PCA-FCN	0.0328	0.0453	0.0383
		RDP-LFN	0.0157	0.0264	0.0252

However, when the network depth is reduced or the number of sampling points is increased, RDP-LFN consistently achieves substantially lower training and validation losses, outperforming US-FCN.

For more intuitive performance evaluation, three representative samples were randomly selected from the test set for comparative analysis. Fig. 5 compares the full-wave simulated $|S_{11}|$ curves using CST with structural parameters predicted by both models. The results indicate that the $|S_{11}|$ curves obtained from EM simulation using the parameters predicted by the RDP-LFN model exhibit closer agreement with the ground truth, demonstrating the effectiveness of the proposed method.

To further validate the superiority of the proposed method, two dimensionality reduction techniques, i.e., fast Fourier transform (FFT) and principal component analysis (PCA), have also been evaluated in the study. As shown in Table I, the proposed RDP-LFN clearly outperforms the other methods.

IV. INVERSE DESIGN OF AN OPTICALLY TRANSPARENT AFSR

The proposed RDP-LFN is subsequently applied in the inverse design of an AFSR. The target $|S_{11}|$ curve undergoes adaptive sampling via the RDP, with the extracted critical features serving as inputs to the LFN model to derive the desired AFSR structure. As shown in Fig. 6, the target $|S_{11}|$ performance is represented by piecewise linear curves defined by the operational bands and resonance points. The absorption bands span 4 GHz to 12 GHz and 24 GHz to 29 GHz, while the reflection band lies between 14 GHz to 22 GHz, with the center frequency of each band set as a resonance point. Fig. 6 also presents a comparison between

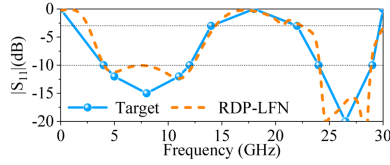


Fig. 6. Comparison between the target $|S_{11}|$ and the simulated $|S_{11}|$ of the inversely designed AFSR.

TABLE II
PERFORMANCE COMPARISON WITH OTHER RELATED WORKS

Ref.	AB (GHz)	RB (GHz)	FBW (%)	TH (λ_L)	RL (dB)	OT
[4]	2.1-4.2, 6.1-9.6	4.3-5.7	127.5	0.084	0.33	×
[7]	4.8-8.0, 10.2-16.0	8.2-9.8	107.7	0.104	—	×
[8]	1.7-3.8, 5.7-8.1	4.3-5.1	130.5	0.064	0.17	×
[9]	0.95-1.34, 2.1-3.5	1.91	114.8	0.101	0.15	×
[10]	2.6-6.9, 7.8-10.0	7.4	117.5	0.118	0.1	×
*	4.1-12.3, 24.1-28.9	14.4-22.4	150.8	0.079	0.22	✓

* = This work; AB = Absorption bandwidth; RB = Reflection bandwidth; FBW = Fractional bandwidth; TH = Thickness; RL = Reflection loss; OT = Optical transparency

the target response and the predicted response of the inversely designed AFSR structure.

While minor frequency shifts are observed in specific bands, the overall frequency response maintains excellent agreement, confirming the RDP-LFN model's capability to accurately map frequency-domain characteristics to structural parameters. The parameters are derived as follows in unit of mm: $h_2 = 5.5$, $g_2 = 0.25$, $l_1 = 2.6$, $l_2 = 1$, $w_1 = 0.39$, and $w_2 = 0.25$.

The generated structure demonstrates $|S_{11}|$ below -10 dB within both 4.05 GHz to 12.3 GHz and 24.1 GHz to 28.9 GHz. According to the formula of absorption $A = (1 - |S_{11}|^2 - |S_{21}|^2) \times 100\%$, the absorption rate exceeds 90% in these bands. Notably, the AFSR exhibits high-efficiency reflection characteristics in the 14.4 GHz to 22.4 GHz range, with a reflection loss of approximately 0.22 dB at 17.1 GHz. The overall electrical thickness of the inverse-designed AFSR structure is only $0.079\lambda_L$ (where λ_L represents the wavelength at the lowest cut-off frequency).

A performance comparison between the proposed AFSR structure and other works is summarized in Table II. The proposed architecture demonstrates superior performance metrics in terms of operational bandwidth and structural thickness. Remarkably, it maintains low reflection loss while preserving optical transparency, exhibiting outstanding comprehensive performance.

V. EXPERIMENTAL VALIDATION

To further validate the performance of the inversely designed AFSR using RDP-LFN, a $300 \text{ mm} \times 300 \text{ mm}$ prototype comprising 30×30 unit cells was fabricated as shown in Fig. 7(a). The optical transmittances of the ITO film and PMMA reported in the literature are approximately 85% [37] and 90% [38], respectively. The multilayer structure was assembled using nylon screws and nuts.

Fig. 7(b) shows the experiment setup, where horn antennas and an Agilent E8363C vector network analyzer are employed to measure $|S_{11}|$ and $|S_{21}|$. The measurement results, depicted

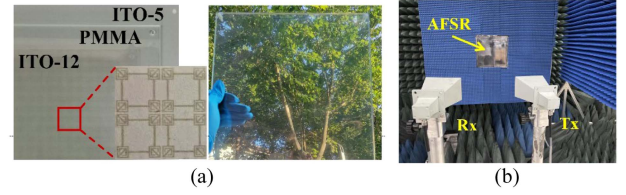


Fig. 7. (a) Photograph of fabricated AFSR and (b) experimental setup.

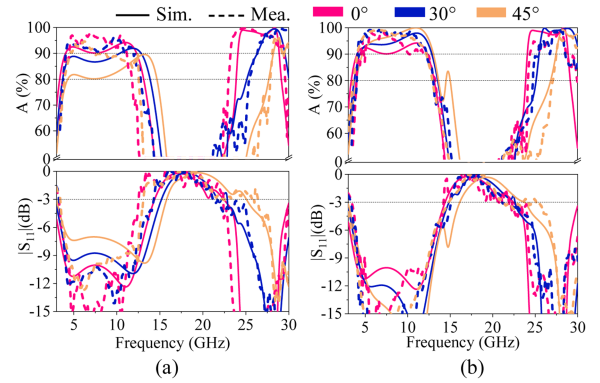


Fig. 8. Comparison of the AFSR performance between measurement and simulation for (a) TE and (b) TM polarizations.

in Fig. 8, are compared to the simulations. Experimental results demonstrate that under normal incidence, the proposed AFSR achieves an absorption rate exceeding 90% within 4.5 GHz to 11.1 GHz and 24.4 GHz to 28.6 GHz, while maintaining a reflection coefficient greater than -3 dB in the 14.3 GHz to 21.9 GHz range. When the incidence angle increases to 45° , the structure maintains an absorption rate above 80% across 3.9 GHz to 12.5 GHz and 28.1 GHz to 29.1 GHz bands for both polarizations, with reflection coefficients surpassing -3 dB in the 14.9 GHz to 21.2 GHz range. Although fabrication tolerances lead to a frequency shift in the TE-polarized absorption band, the measured $|S_{11}|$ response shows good agreement with the overall simulated results, thereby validating the effectiveness of the proposed methodology.

VI. CONCLUSION

This study proposes an RDP-LFN inverse design framework to address the challenges of high-dimensional data and feature redundancy in conventional US. The RDP-LFN consistently achieves lower loss compared to US-FCN methods with various sampled points and FCN depths. Based on this framework, the inverse design of a single-layer optically transparent UWB AFSR has successfully been achieved, as a paradigm. The designed structure demonstrates absorption exceeding 90% in the 4.05 GHz to 12.3 GHz and 24.1 GHz to 28.9 GHz, while maintaining $|S_{11}|$ above -3 dB in the 14.4 GHz to 22.4 GHz, with good agreement between the measured and simulated results. This is the first attempt to incorporate non-uniform sampling into NN-assisted FSS design, establishing a foundation in advancing intelligent design of EM devices.

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